

Ceed 2024 Complete solution

Common Entrance Examination for Design

CEED 2024

Question Paper

INSTRUCTIONS

1. The total duration of the examination is 3 hours. The question paper contains two parts - Part A and Part B. The duration of Part A is one hour. The duration of Part B is two hours. Part B will start only after Part A ends. Part A cannot be attempted once Part B commences.
2. Questions of Part A and Part B will appear on the computer. Answers to **Part A** have to be entered in the **computer**. Answers to **Part B** need to be given in the **answer booklet** provided by the invigilator.
3. Part A carries a total of 150 marks. It contains 3 sections. All questions are mandatory. The following table summarises the marking scheme of Part A:

Part A Section	Type of questions	Number of questions	Marks for each correct answer	Marks for each wrong answer	Marks for each question not attempted	Total marks for the section
I	Numerical Answer Type (NAT)	8	4	0	0	32
II	Multiple Select Question (MSQ)	10	Partial Marking	-1	0	40
III	Multiple Choice Question (MCQ)	26	3	-0.5	0	78

4. For each NAT question, the answer is a **number**. The answer needs to be entered using the virtual keyboard on the monitor. No choices will be shown for these questions. In NATs, the correct answer will be awarded **4 marks**. There is no negative or partial marking for NATs.
5. Each MSQ has four choices of which **one or more** is/are the correct answer(s).

The following is the marking scheme:

Full Marks : + 4 If only (all) the correct option(s) is(are) chosen and NONE of the incorrect options are chosen

Partial Marks : + 3 If all the four options are correct but ONLY three options are chosen

Partial Marks : + 2 If three or more options are correct but ONLY two options are chosen, both of which are correct and NONE of the incorrect options are chosen

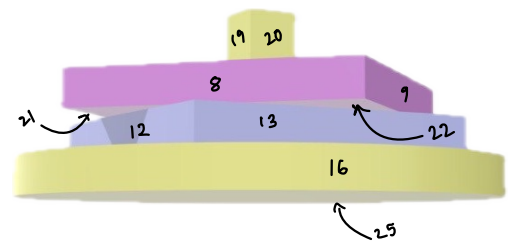
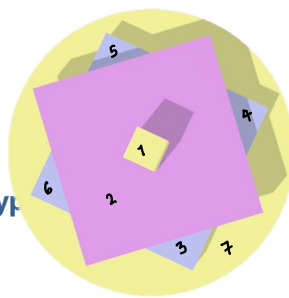
Partial Marks : + 1 *If two or more options are correct but ONLY one option is chosen and it is a correct option and NONE of the incorrect options are chosen*

Zero Marks : 0 *If none of the options is chosen (i.e., the question is unanswered)*

Negative Marks: -1 *In all other cases*

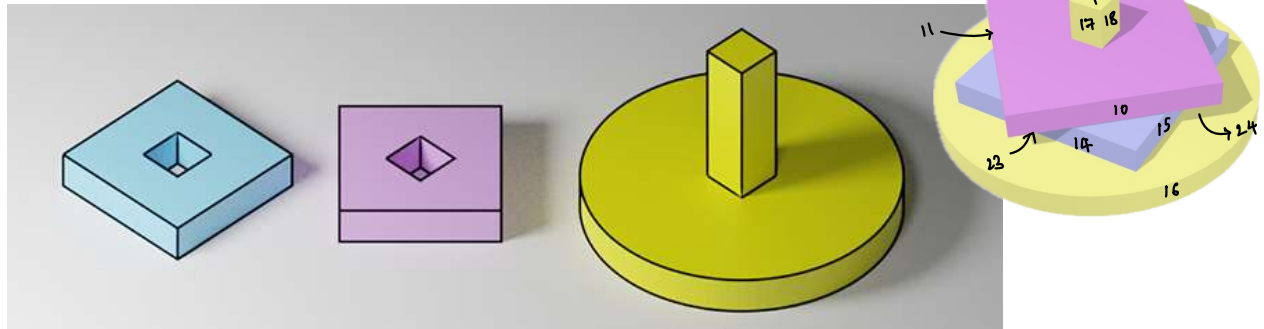
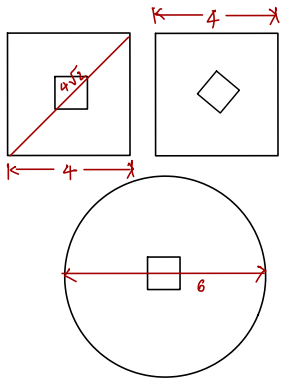
6. Each MCQ has four choices of which **only one** is the correct answer. In MCQs, the correct answer will be awarded **3 marks** and wrong answer will be awarded **-0.5 marks**. Questions not attempted will be awarded zero marks.
7. Scribble pads will be provided for rough work for Part A. These have to be returned to the invigilator at the **end of Part A exam**.
8. Part B carries a total of 100 marks. It contains 5 questions of 20 marks each. **All questions** in Part B are **mandatory**. Each question must be answered on the page(s) designated for that question in the answer booklet. Additional instructions to Part B questions are provided in the answer booklet.
9. In Part B, colour pencils, crayons, sketch pens etc. may be used unless otherwise specified in the questions.
10. Charts, graph sheets, tracing papers, tables, calculators, cellular phones and other electronic gadgets are not allowed in the examination hall.

Part A – Section I: Numerical Answer Type



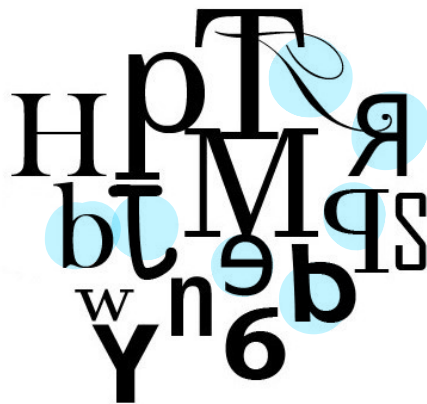
- Q.01** The two objects on the left have a height of 1 cm and sides of 4 cm. The yellow object on the right has a total height of 5 cm. The cylindrical base of the yellow object has a diameter of 6 cm and a height of 1 cm. The two objects on the left are to be stacked on the yellow object on the right. After all the objects are stacked together, how many outer surfaces will the resulting structure have?

Ans 25



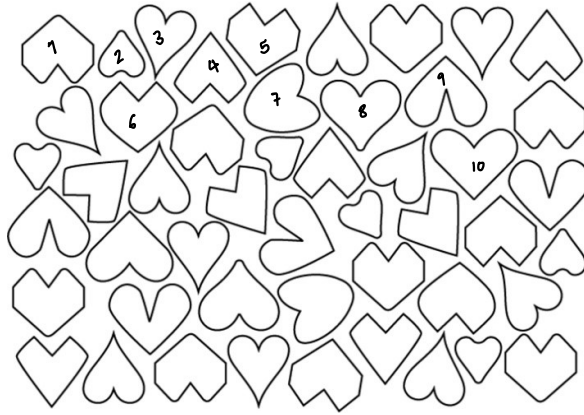
- Q.02** A "group of letters" has a few letters flipped on vertical axis. The mirror image of this "group of letters" is shown below. What is the total number of flipped letters in the group?

Ans 7



J, R, P, e, d, J, d

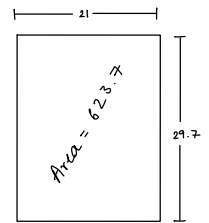
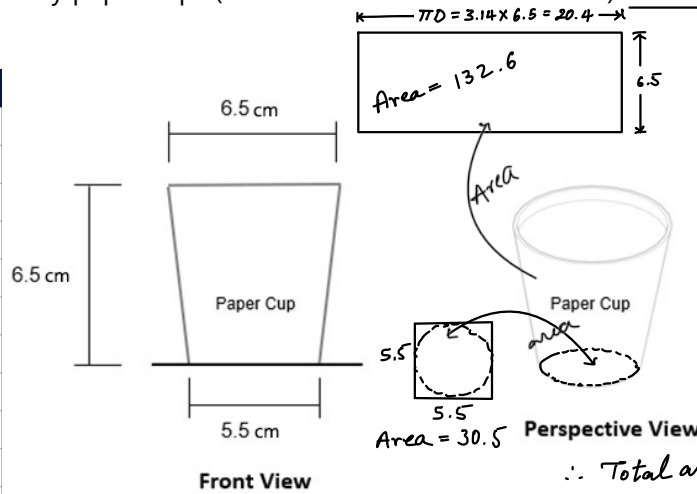
- Q.03** How many unique shapes are there in the image below?



Ans 10

Q.04 How many paper cups (of the dimensions shown below) can be made from one A4 paper? *Ans 3*

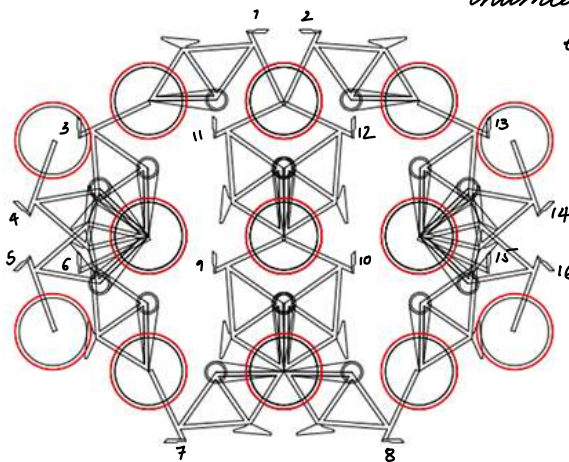
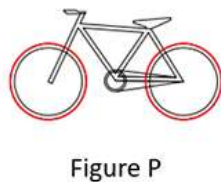
Paper Size	Millimetres	Centimetres	Inches
A0	841 x 1189mm	84.1 x 118.9cm	33.1 x 46.8in
A1	594 x 841mm	59.4 x 84.1cm	23.4 x 33.1in
A2	420 x 594mm	42 x 59.4cm	16.5 x 23.4in
A3	297 x 420mm	29.7 x 42cm	11.7 x 16.5in
A4	210 x 297mm	21 x 29.7cm	8.3 x 11.7in
A5	148.5 x 210mm	14.85 x 21cm	5.8 x 8.3in
A6	105 x 148.5	10.5 x 14.85cm	4.1 x 5.8in
A7	74 x 105mm	7.4 x 10.5cm	2.9 x 4.1in
A8	52 x 74mm	5.2 x 7.4cm	2.0 x 2.9in
A9	37 x 52mm	3.7 x 5.2cm	1.5 x 2.0in
A10	26 x 37mm	2.6 x 3.7cm	1.0 x 1.5in



$\therefore$ Total no. of cups that can be made is $\frac{623.7}{163.1} \approx 3$

$$\therefore \text{Total area} = 132.6 + 30.5 = 163.1$$

Q.05 Figure P is the image of a complete bicycle. If parts can be shared between two or more bicycles and flipping of Figure P is allowed, how many complete bicycles are there in the Figure Q?



number of handles is equal to the number of Bicycles.

$\therefore$ Ans 16.

Figure Q

Ans 162

Q.06 What number will replace the question mark?

2 4	10 5	12 9
12	75	? → 162

6 7	8 3
63	36

$$2 \times 4 = 8 + \frac{8}{2} = 12$$

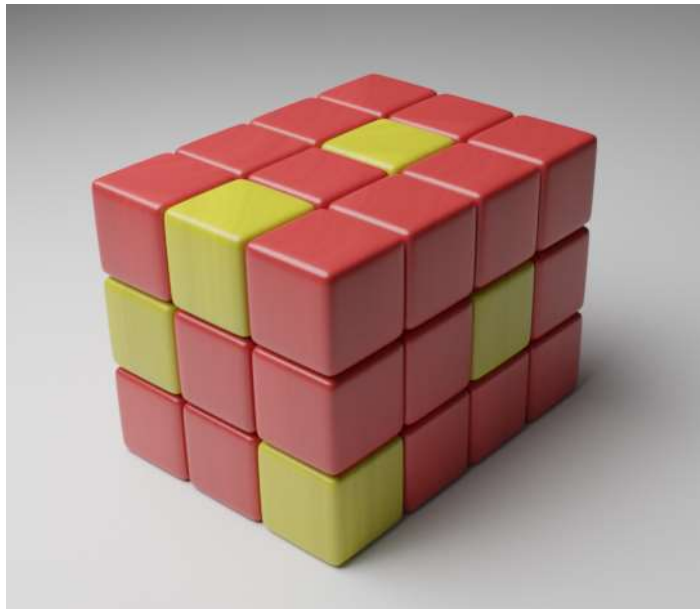
$$10 \times 5 = 50 + \frac{50}{2} = 75$$

$$12 \times 9 = 108 + \frac{108}{2} = 162$$

$$6 \times 7 = 42 + \frac{42}{2} = 63$$

$$8 \times 3 = 24 + \frac{24}{2} = 36$$

Q.07 Coloured cubes (red and yellow) of equal size are arranged and stacked to form the cuboidal structure as shown in the image. If the ratio of the number of yellow cubes to that of the red cubes is 1:3, how many yellow cubes are NOT visible in the image?



$$Y : R = 1 : 3$$

$$\text{Total number of Small cubes} = 4 \times 3 \times 3 = 36$$

∴

$$Y + R = 36 \quad \therefore \frac{36}{4} = 9$$

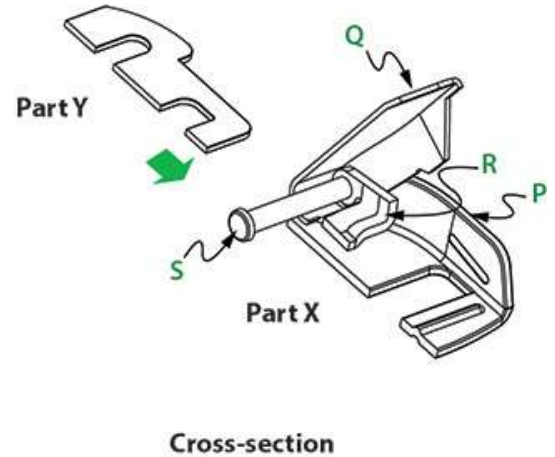
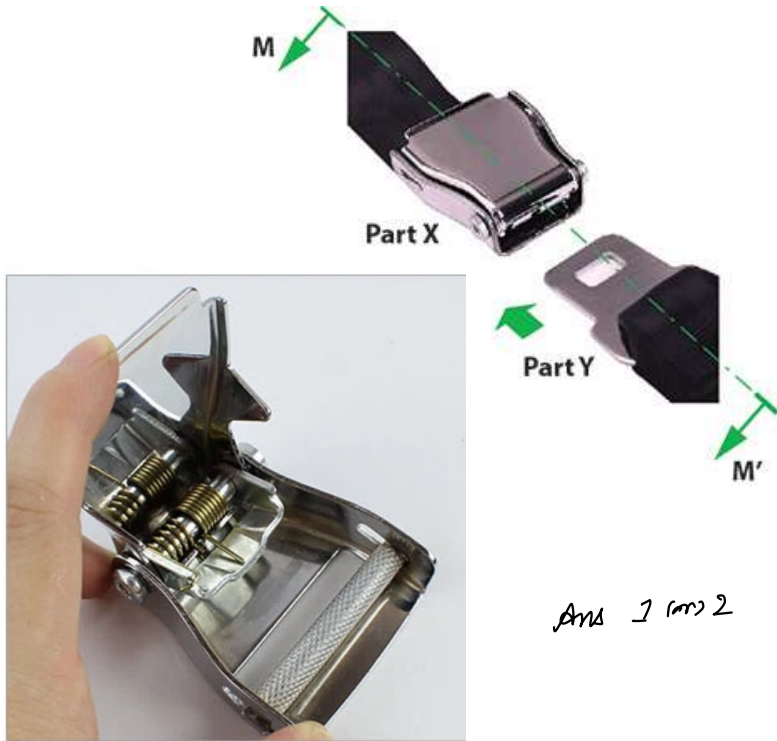
$$Y : R = 1 : 3$$

$$\begin{array}{r} \times 9 \quad \times 9 \\ \hline 9 \quad 27 \end{array}$$

∴ There are "9" yellow small cubes

and visible cubes are 5
∴ 9 - 5 = 4

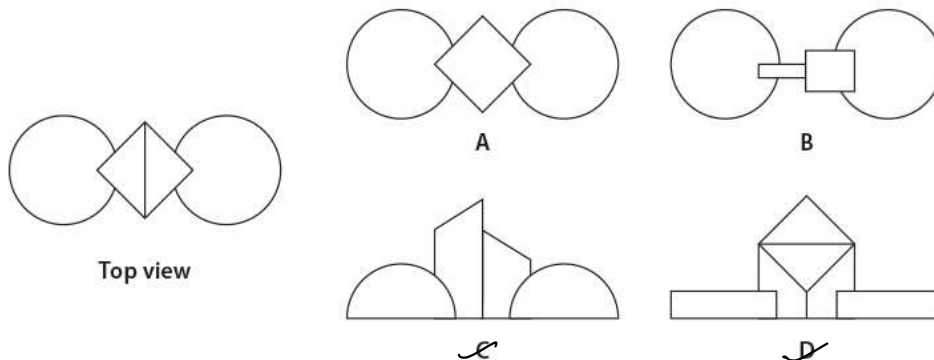
Q.08 Shown below is the view and cross-section of a seat belt used in aeroplanes. Part X has four different components P, Q, R and S. How many springs are required to be fixed on the part S to make the seat belt work?



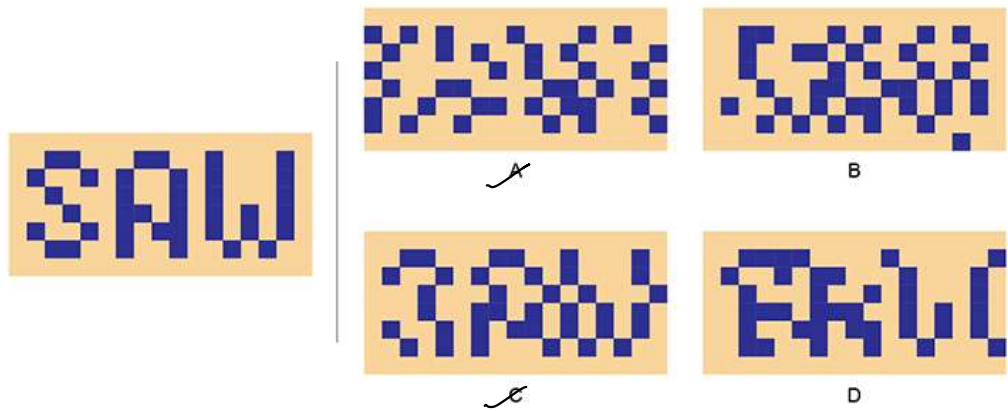
Ans 1 and 2

Part A – Section II: Multiple Select Questions

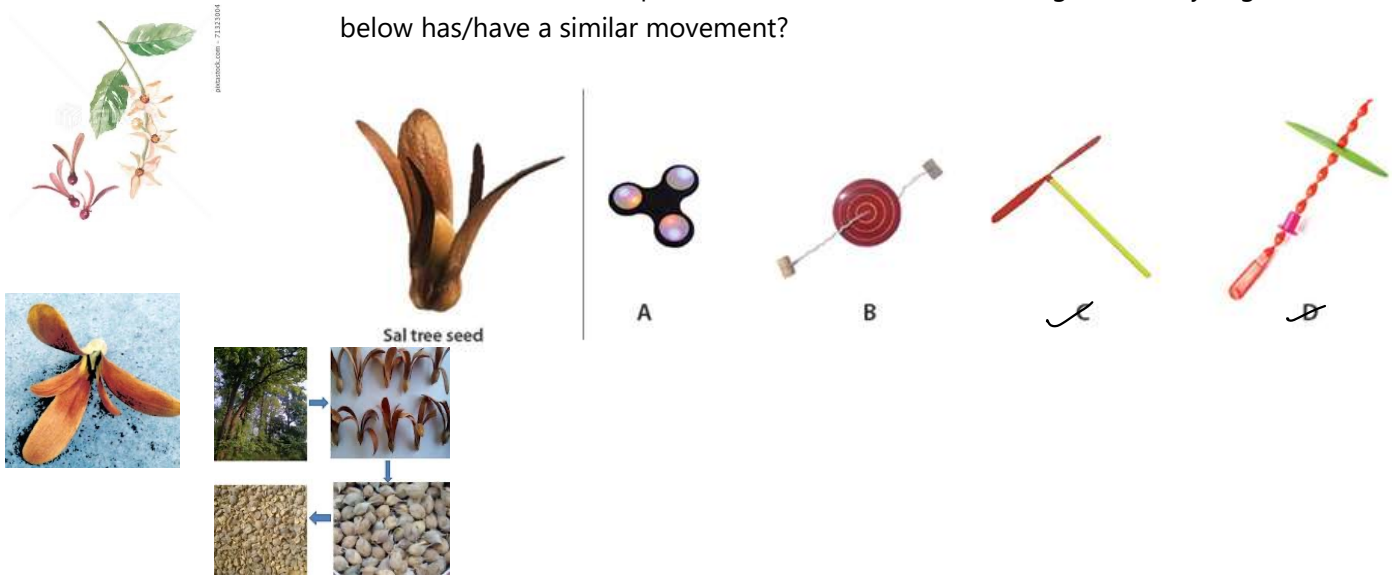
Q.09 Shown below (on the left) is the top view of a 3D form. Identify the possible front view(s). Ans C, D



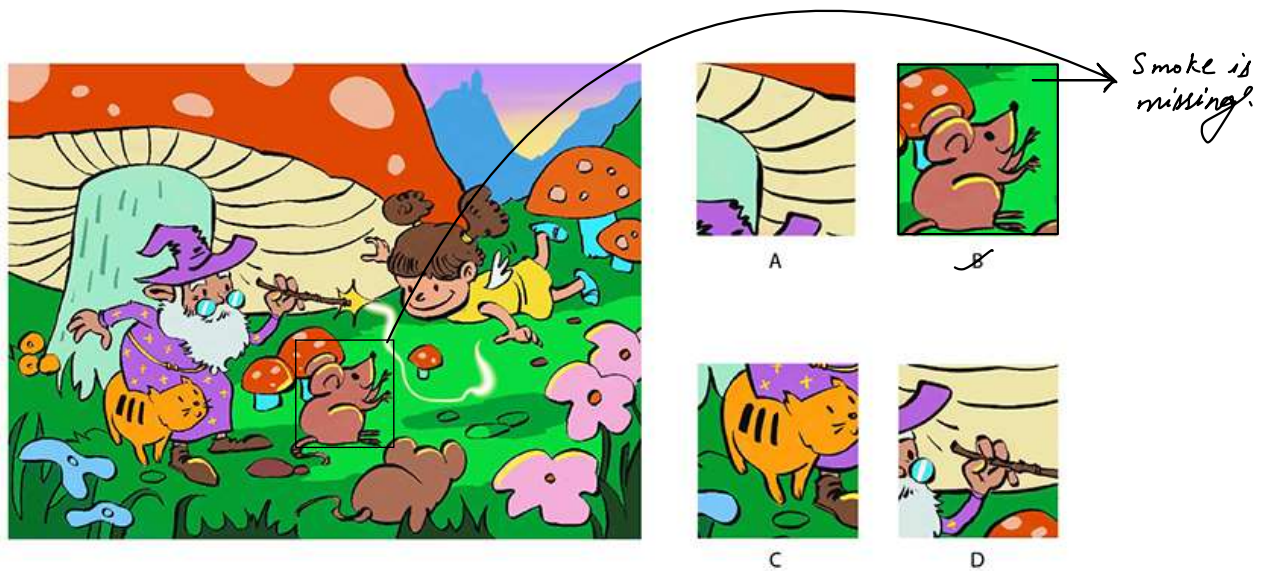
Q.10 The pixels in the image on the left are shifted horizontally to create one or more options on the right side. Identify the correct option(s).



Q.11 Sal tree seed moves in a particular manner, while free falling. Which toy(s) given below has/have a similar movement?



Q.12 Which option(s) is/are NOT a part of the image on the left?



- Q.13** A professional animation studio has created a paper cut-out model with movable joints as shown below. This is to be used for an animation feature film. Which material(s) will the studio use for the joints?



- ☒ A. Nylon Wire
- ☐ B. Mouldable Clay
- ☒ C. Eyelet Rivet
- ☐ D. Chalk

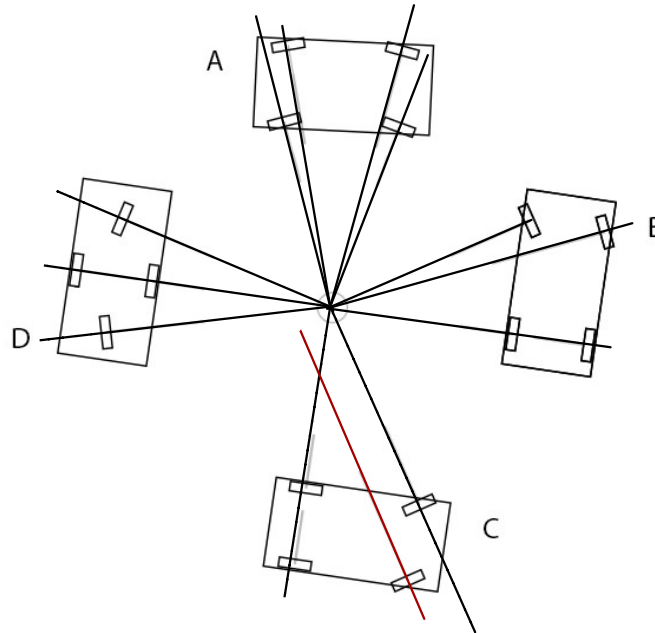


- Q.14** Which of the materials given in the options is/are used in the painting style shown below? (*wrong options*)



- A. Brick
- B. Vegetables
- C. Minerals
- D. Gold

Q.15 Four-wheeler vehicles are trying to take a circular path around the marked point in the centre. Select the vehicle(s) with the correct wheel positions. *Ans A B D*



Q.16 9 coloured tiles are arranged in a 3x3 columns and rows as follows. *Ans B, C*

1. Red, Blue and Purple are in the corners.

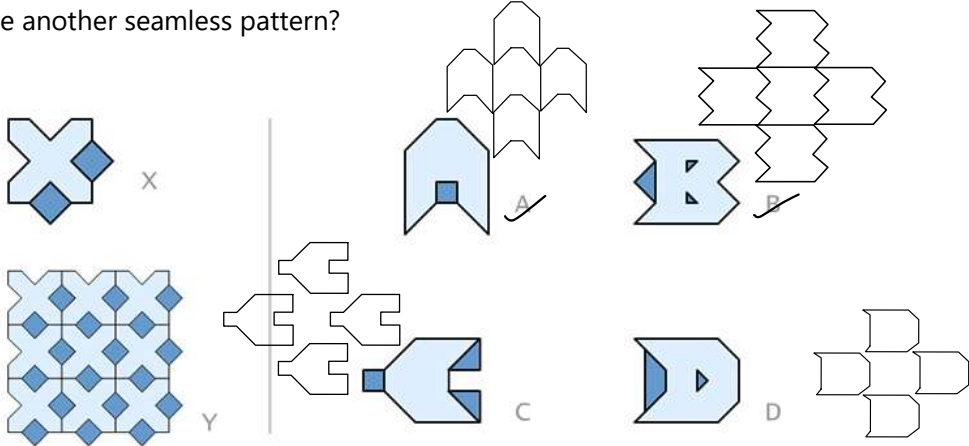
Purple	Pink	Blue
Orange	Yellow	Black
Red	Brown	Green

- Brown, Pink and Yellow are arranged linearly.
- Black touches Green and Blue.
- Orange does not touch Brown and Pink.
- Blue touches Pink.
- Red touches Brown.

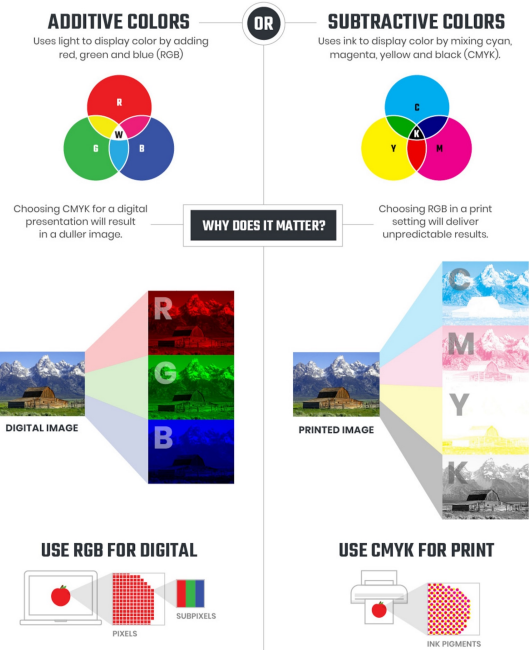
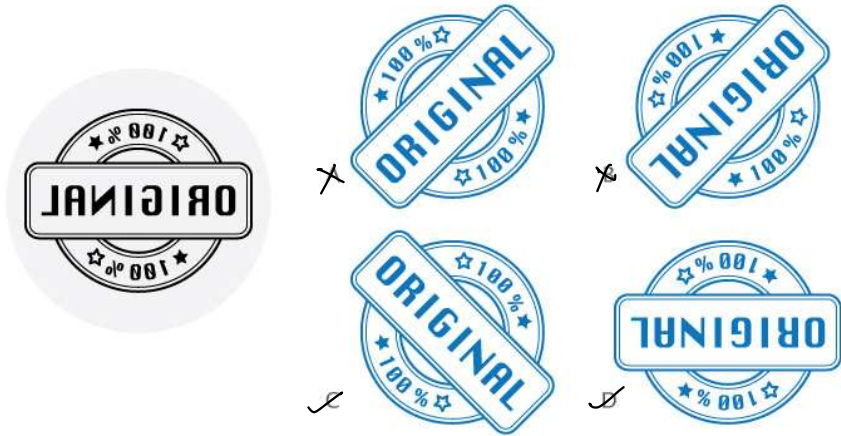
Which of the statements is/are TRUE?

- A. Additive primary colours are in diagonal arrangement.
- ☒ B. Red and Green is either in the same row or column.
- ☒ C. Purple and Green are diagonally opposite to each other.
- D. Pink is in the centre.

Q.17 Tile X was used to create a pattern which is seamless (without gaps and/or overlaps) when arranged as shown in Y. In a similar manner, which tile(s) from the options will *Ans A,B* create another seamless pattern?

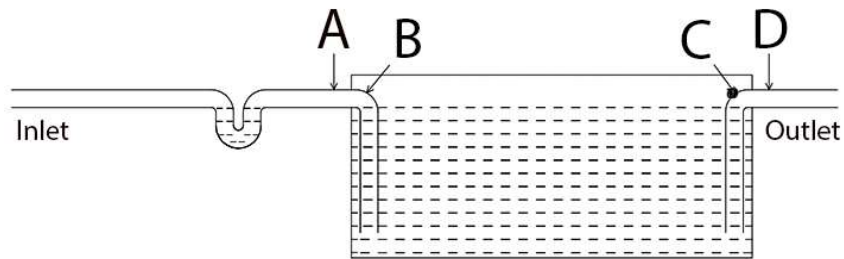


Q.18 Which of the option(s) is/are created by the stamp shown on the left? *Ans C D*

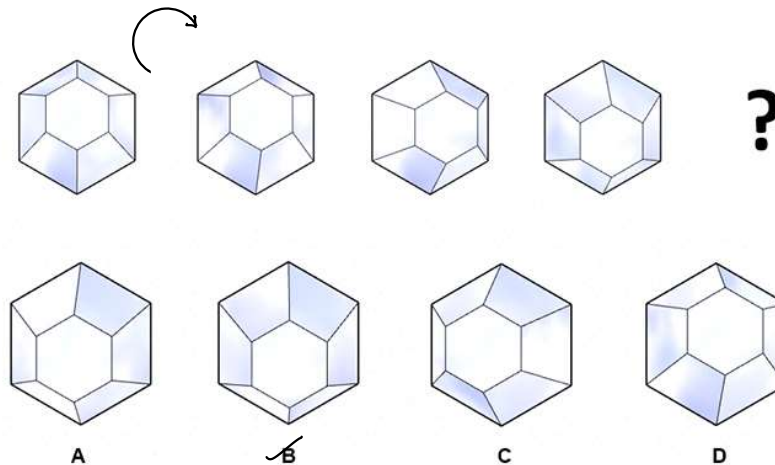


Part A – Section III: Multiple Choice Questions

- Q.19** A tank is converted to treat organic waste water from kitchen as shown in the figure below. Bio gas gets accumulated on top part of the tank. Which of the following options is a suitable location for drilling an air hole to maintain the water level inside the tank? *Ans C*



- Q.20** Which option will replace the question mark? *Ans B*



- Q.21** Parts of the letters have been hidden by vertical strips in the given options. Which option contains a meaningful word with correct spelling? *Ans D*

ROD GET EOTME

A

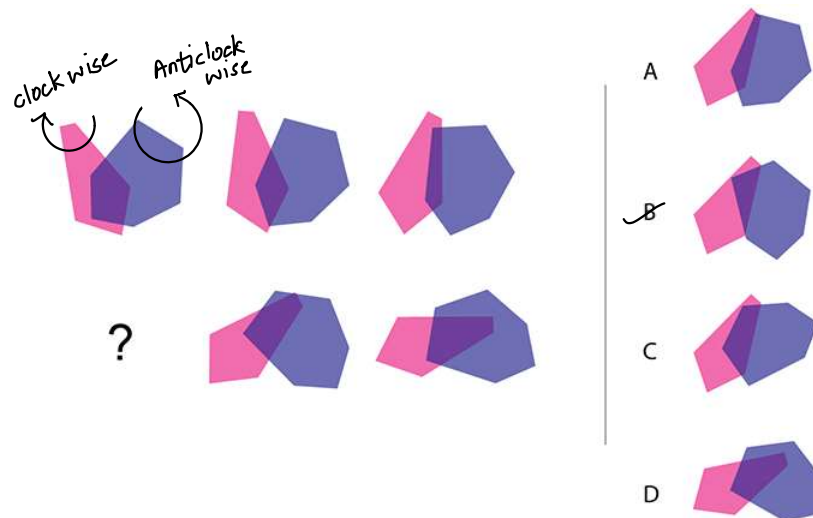
B

FO'C'E ROTATE

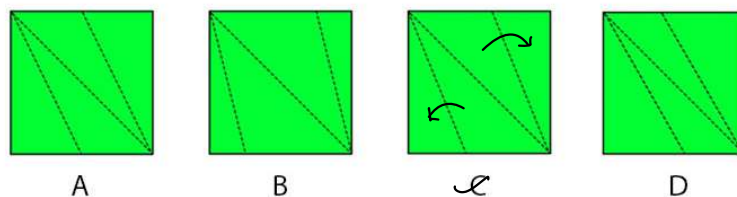
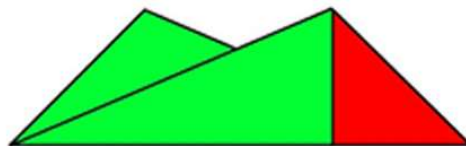
C

D

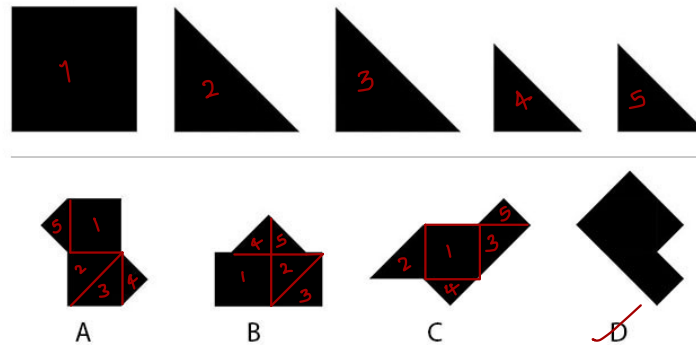
Q. 22 Which option will replace the question mark? *Ans B*



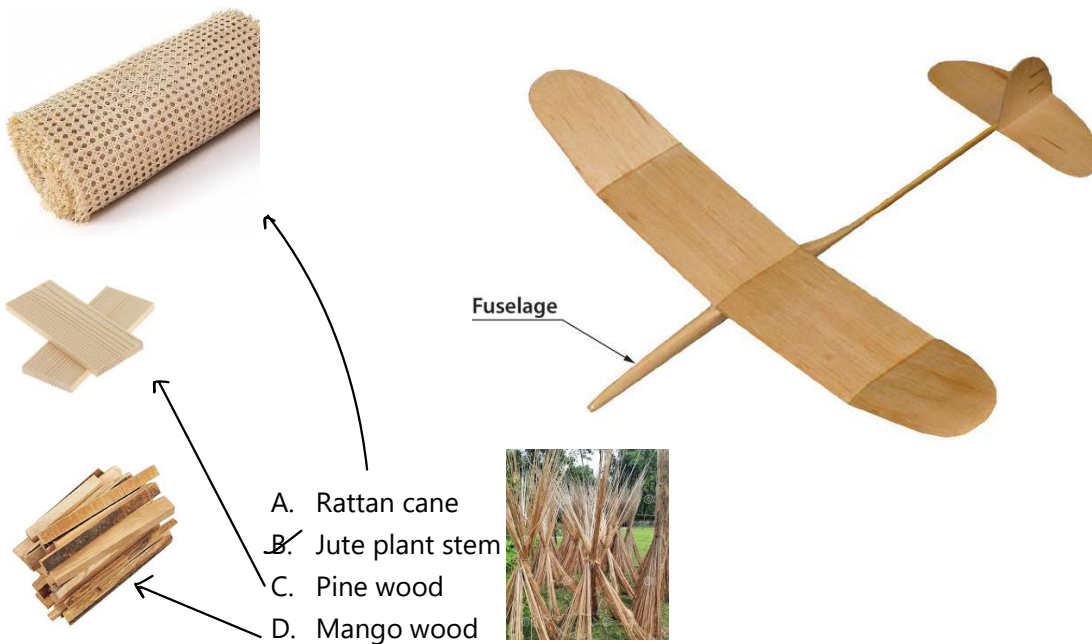
Q.23 Given is the one side of a folded sheet of paper with green color on one side and red color on the other side. Find the correct option when this sheet is unfolded. Dotted lines represent the fold lines. *Ans C*



- Q.24** One square and 4 triangles given below are used to make shapes without overlapping. Among the options, which is the shape that CANNOT be made? *Ans D*

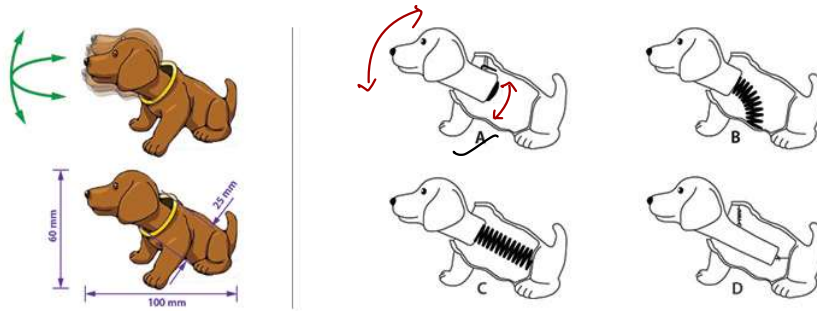


- Q.25** A glider toy as shown in the image below is usually made with Balsa wood. Due to the unavailability of Balsa wood, a student is considering an alternative material. Considering the density of the material that is required to make the fuselage, which will be the most suitable option?

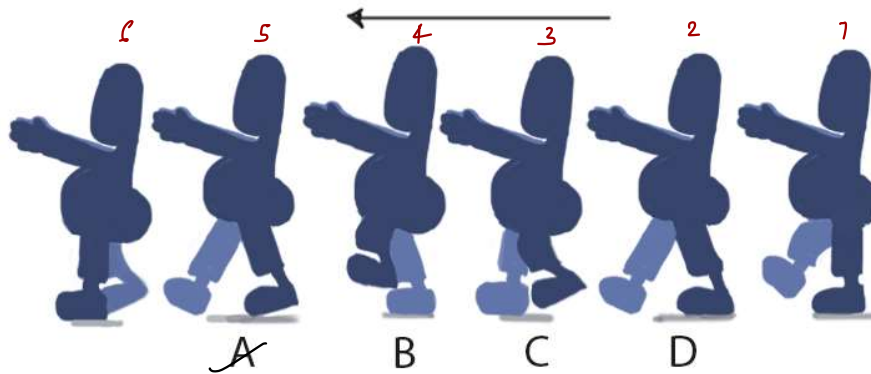


- Q.26** What is the internal mechanism for producing the wobbling head movement in the toy shown below? Assume that the toy needs to be mass manufactured.

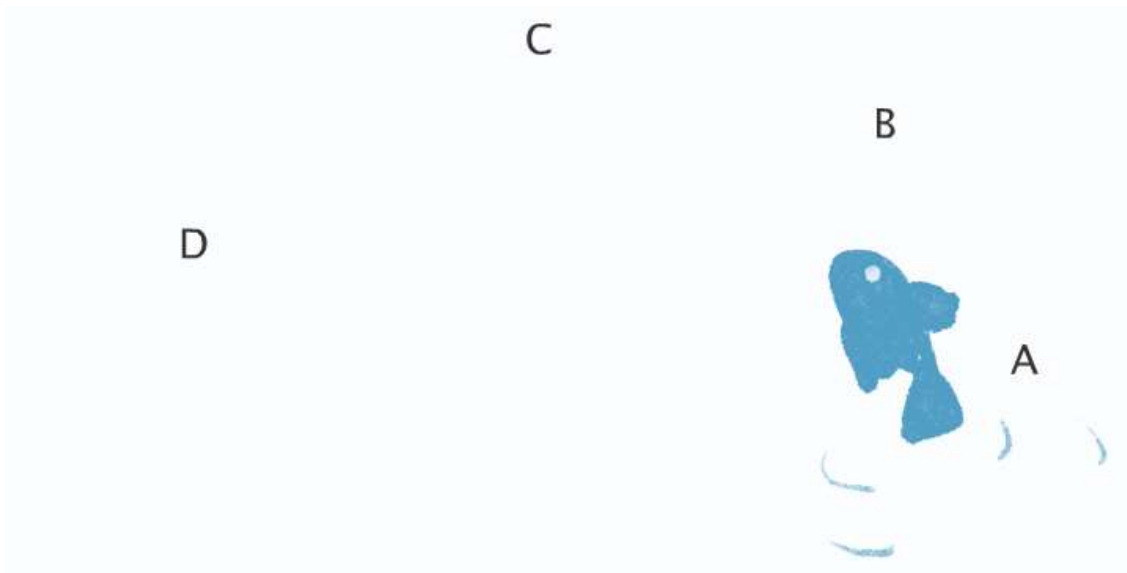
Ans A



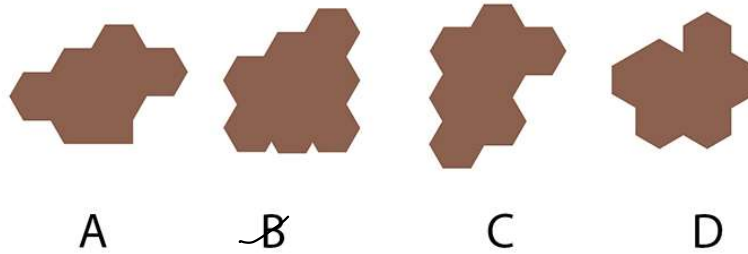
Q.27 A sequence of a man sleep walking is shown below. Which frame is incorrect? *Ans A*



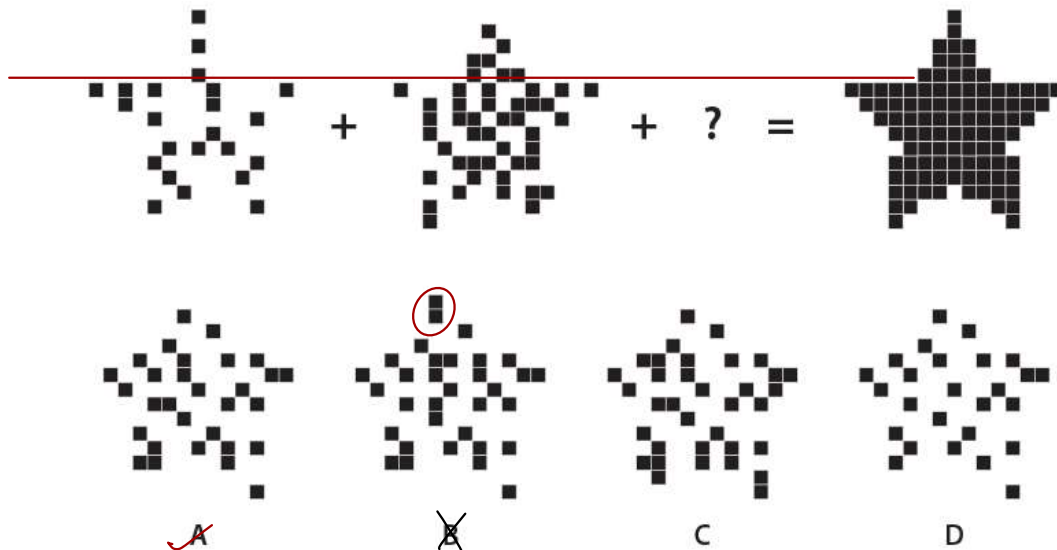
Q.28 Shown below is an animation of a fish jumping. At which point, we need to draw maximum number of frames?



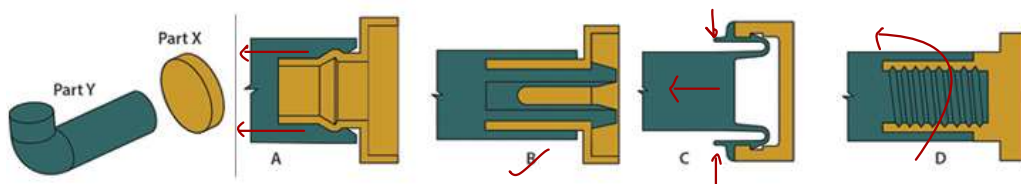
Q.29 Which among the following shapes has the largest area?



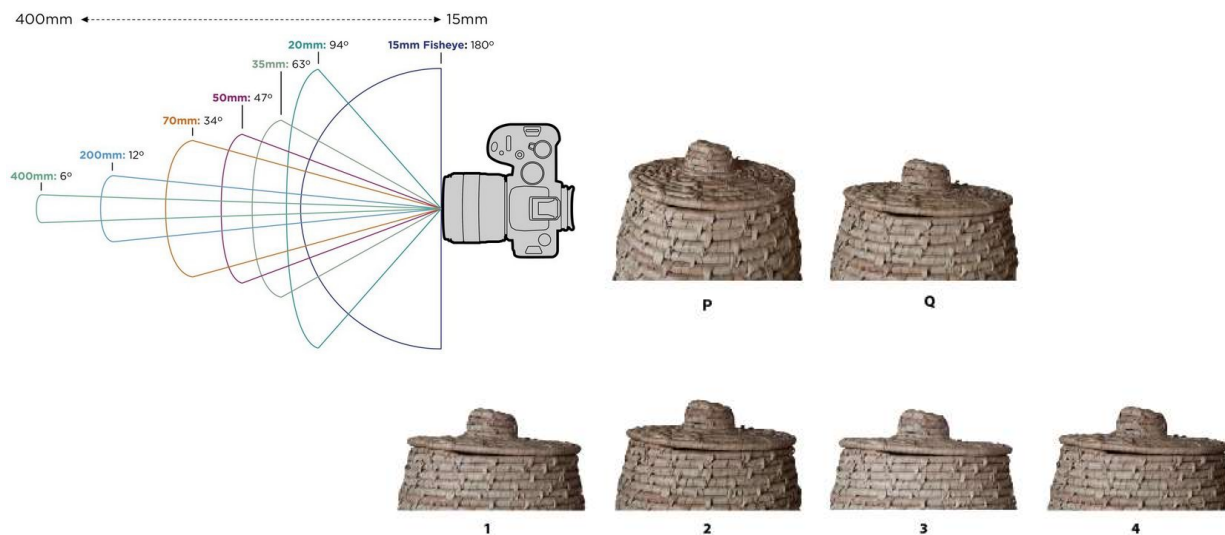
Q.30 Which option will replace the question mark?



Q.31 Part X and part Y are to be assembled together to make a hanger in such a way that it can be disassembled, if required. Which of the following is not an appropriate option to join part X and part Y?

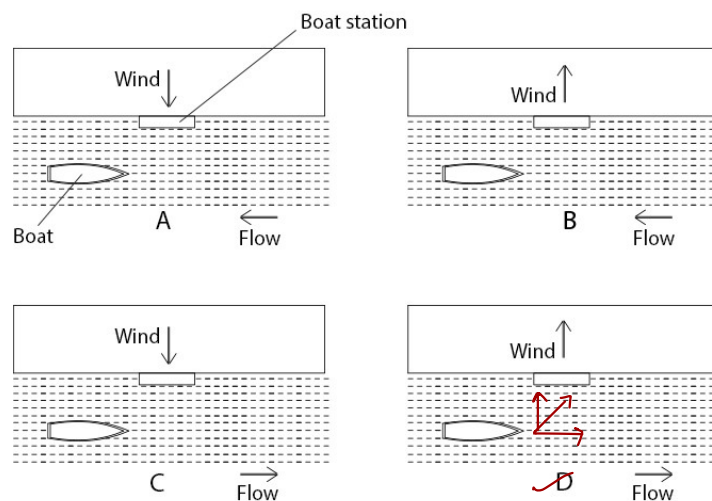


Q.32 Q is shot with higher focal length than P. Arrange 1, 2, 3, 4 in the ascending order of the focal length they are shot with.

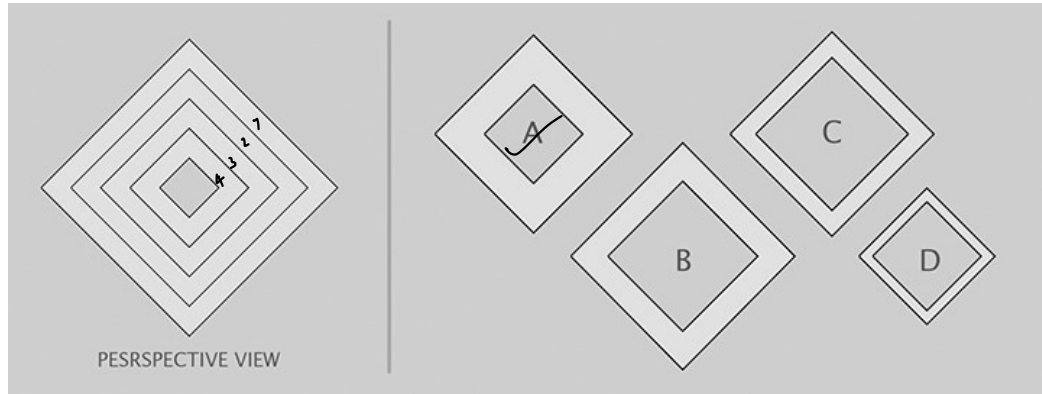


- A. 4, 1, 3, 2
- B. 2, 1, 4, 3
- C. 4, 3, 1, 2
- ☒ D. 2, 4, 1, 3

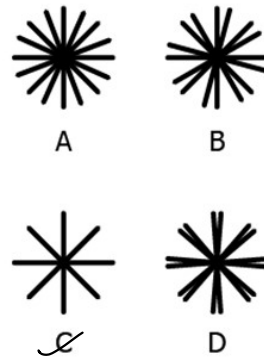
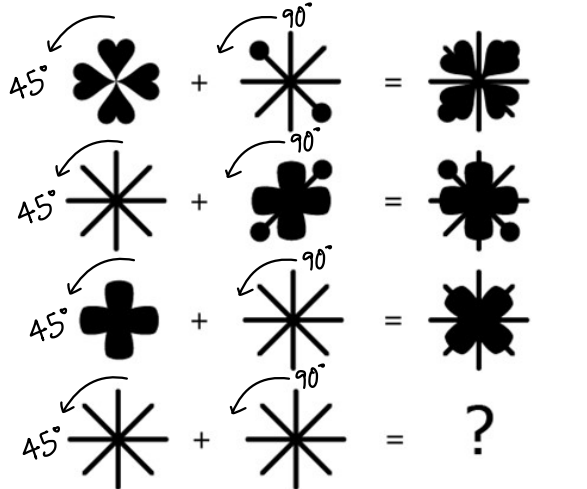
Q.33 Identify the most ideal condition for the boat to approach the station. *Ans D*



Q.34 A perspective view of four frames spaced from near to far at equal intervals, is seen on the left. The frames perfectly align without any overlaps in this view. The options show the same frames arranged side by side on a single plane. Which of the frames in the options is the farthest from the viewer in the perspective view?



Q.35 Which option will replace the question mark?



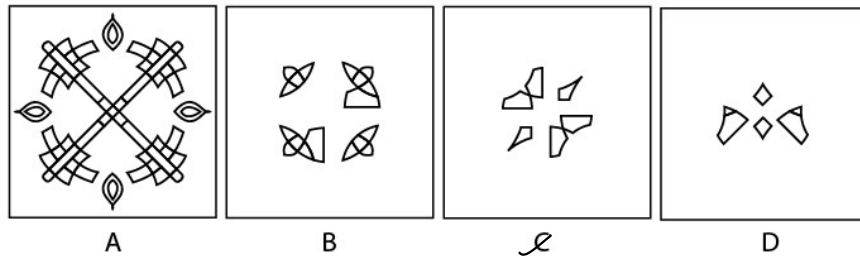
Q.36 Which option is correct for all the signs given below (from left to right)? *Ans B*



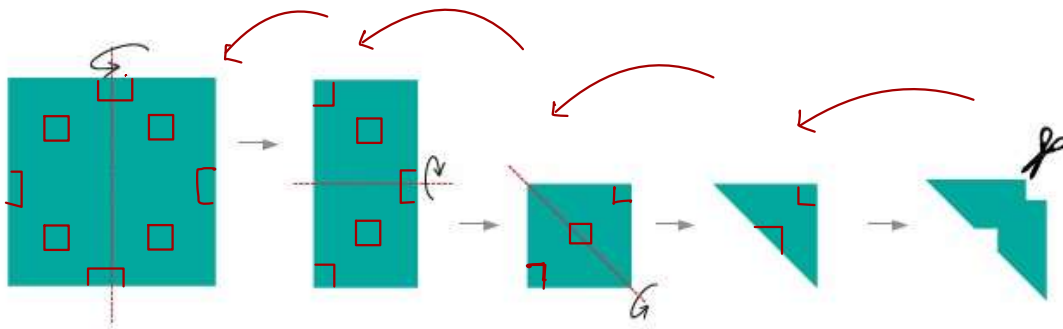
- A. No Entry; Stop; Height limit; Keep left and straight; Prohibited road
- ☒ B. Give a way; No entry; Height limit; No overtaking; No stopping
- C. Give a way; No entry; Keep Distance; No overtaking; Prohibited road
- D. No entry; Give a way; Width limit; Keep left and straight; No stopping



Q.37 Which option cannot be used to create the given image below? The options may be rotated. *Ans C*

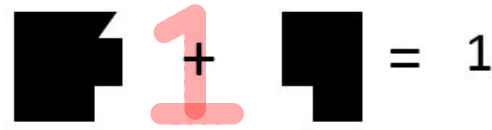


Q.38 A piece of paper is folded and cut as shown below. Which is the correct option, when the paper is unfolded?



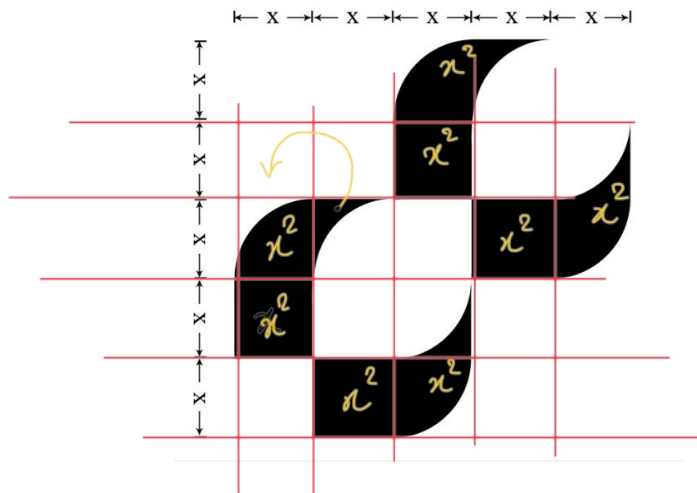
Q.39 Which option will replace the question mark?

Ans B



- A. 4
- ✓ B. 7
- C. 8
- D. 5

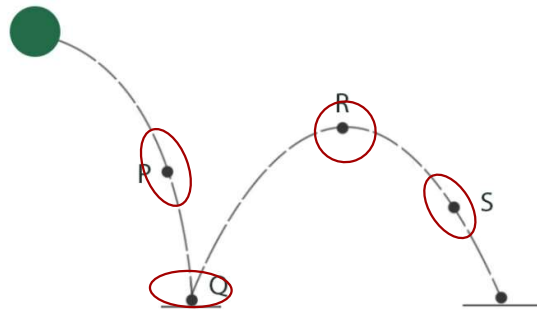
Q.40 What is the total area of the black image below?



- A. $16X$
- B. $4X^2$
- C. $2X^4$
- ✓ D. $8X^2$

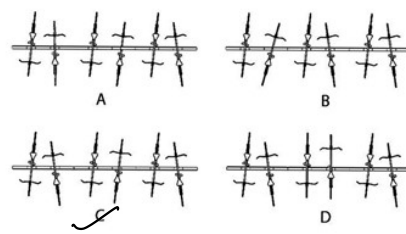
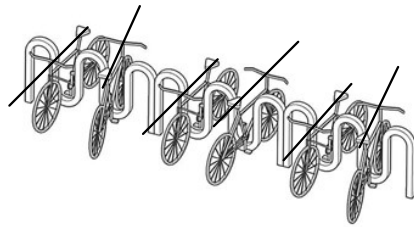
$$\begin{aligned} \therefore \text{Ans} &= x^2 + x^2 + x^2 + x^2 \\ &\quad + x^2 + x^2 + x^2 + x^2 \\ &= 8x^2 \end{aligned}$$

Q.41 An animator is animating a bouncing ball along the dotted path as shown below. Which option is FALSE?



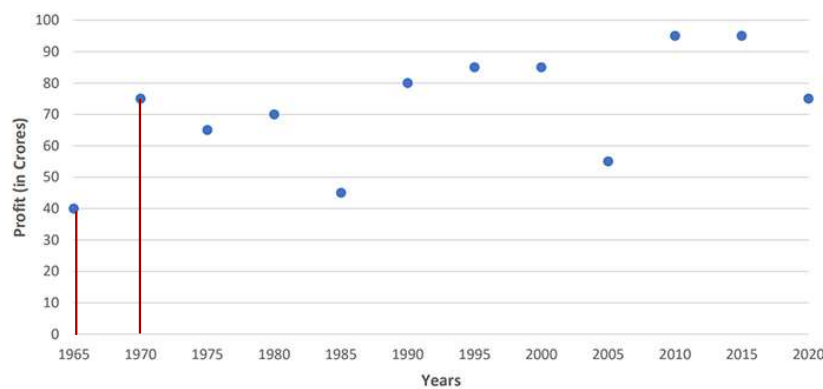
- A. In position P, the ball will stretch vertically.
- B. In position R, the ball will remain the same.
- C. In position Q, ball will stretch horizontally.
- ~~D.~~ In position S, the ball will remain the same.

Q.42 Which among the following options is the correct top view?



by observing all options first and last Cycle.
The Cycle which seems to be straight is Slant and The Cycle which seems to be slant is straight.

Q.43 The annual profit chart of a company is given below. Which option is the correct statement?

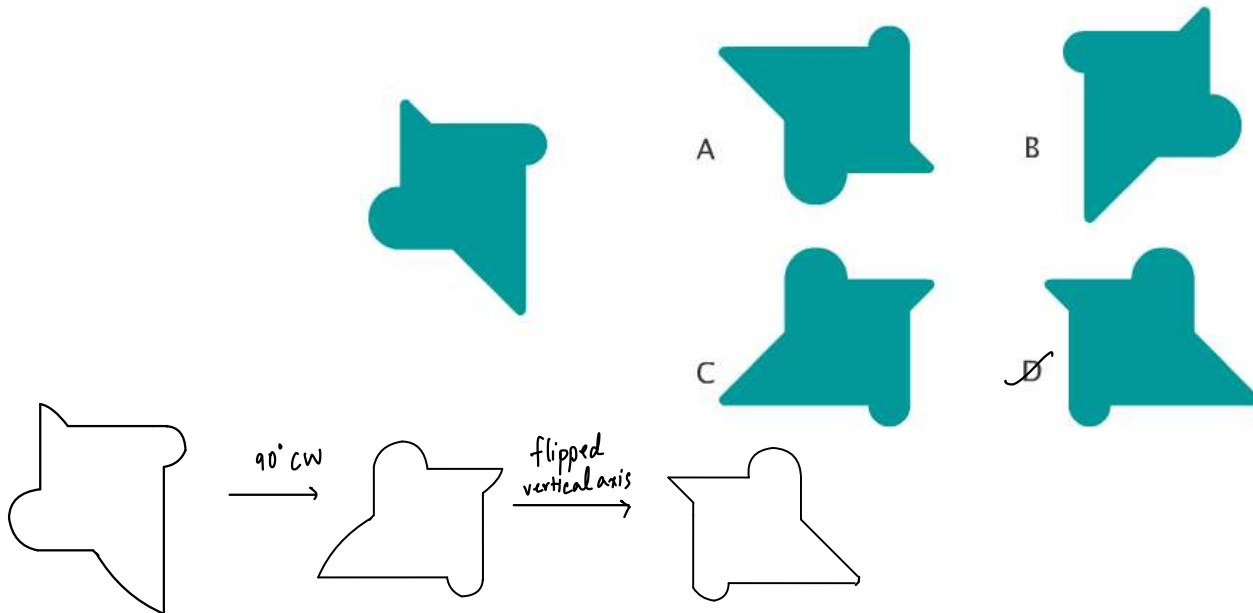


A. 1965 → 40
1970 → 75

- ~~A.~~ The percentage increase between 1965 and 1970 is more than 75%
- B. The percentage decrease between 1985 and 1990 is lesser than 27%
- C. The percentage increase between 2005 and 2010 is more than 100%

D. The percentage decrease between 2015 and 2020 is lesser than 10%

Q.44 The image on the left is rotated 90 degrees clockwise and then flipped along the vertical axis. Which option represents the resulting image?



Part B

All questions are mandatory

Q.1 Sketching (20 marks = 4 x 5 marks)

7-year-old Chinnu has collected all her old toys in a carton. The carton weighs around 2 kg. Draw Chinnu lifting the box and keeping it in the top shelf of her cupboard in the 4 frames given in the answer sheet.

Additional Instructions:

- Use only pencils
- Do not use colours

Evaluation criteria:

- Humour
- Expressions
- Body Language
- Anticipation and exaggeration

Q.2 Creativity (20 marks = 4 x 5 marks)

Imagine the human head as a cube. With this scenario, illustrate the following interactions inside the given boxes (in the answer booklet) in this order:

1. Human-human
2. Human-animal
3. Human-environment
4. Human-object

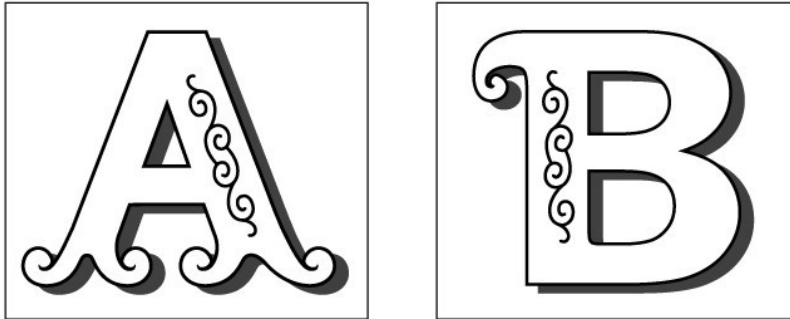
Use only BLACK pencil/pen. Draw only within the boundaries of the boxes. Write the titles below the boxes, in the space provided.

Evaluation criteria:

- Uniqueness: Ability to create unusual images
- Diversity: Variation between each created image
- Clarity: Effective use of visual elements and titles
- Fun: Ability to use humor and surprise in the images and titles

Q.3 Visual Sensitivity (20 marks = 4 x 5 marks)

Shown below are two letters A & B of an ornamental font. Using this font style create the word 'Forest' (the first letter in uppercase and the remaining in lowercase).



Evaluation criteria:

- Consistency in visual characteristics
- Font anatomy
- Proportion
- Rendering skills

Q.4 Form sensitivity (20 marks = 5 x 4 marks)

Inspired from the given three-dimensional form, conceptualise, sketch, and render a single seater furniture for a living room.



Evaluation Criteria:

- Consistency with the visual language of the form
- Form should complement function
- Relevant details of the product
- Surface shading

- Quality of lines/strokes

Q.5 Problem identification (20 marks= 5 + 3 x 5 marks)

Shown below is the image of a rural kitchen. There are many challenges in using such setup for cooking and related activities.



Draw and annotate FIVE distinct design problems in this context.

- Use only one visual for each problem.
- Answer in the given space only.
- Adhere to no more than 5 problems.

Evaluation Criteria:

- Significance of the identified problems
- Ability to identify unique problems
- Ability to explain the problem through visuals

- Quality of lines/strokes

Q.5 Problem identification (20 marks= 5 + 3 x 5 marks)

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